



MASTER THESIS TOPIC

Student: **Bc. Ivana Dukayová**
Student's ID: 115979
Study programme: Automation and Information Engineering in Chemistry and Food Industry
Study field: Cybernetics
Thesis supervisor: Ing. Michaela Horváthová, PhD.
Head of department: prof. Ing. Miroslav Fikar, DrSc.
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Workplace: Oddelenie informatizácie a riadenia procesov (ÚIAM FCHPT)

Topic: **Predictive Control of a Heat Exchanger using the Koopman Model**

Language of thesis: English

Specification of Assignment:

1. Perform experimental identification of the plate heat exchanger and obtain a first-order linear model from measured input–output data.
2. Develop a data-driven Koopman model to capture nonlinear system behavior.
3. Design constrained Model Predictive Control for both models, including state estimation.
4. Analyse and experimentally verify the control performance of both approaches on the heat exchanger.

Length of thesis: 50

Selected bibliography:

1. VALÁBEK, Patrik; HORVÁTHOVÁ, Michaela; PANNOCCHIA, Gabriele; KLAUČO, Martin. Experimental validation of deep Koopman MPC for real-time pasteurization unit control. *Control Engineering Practice*, p. 1–11.
2. KLAUČO, Martin; KVASNICA, Michal. *MPC-Based Reference Governors: Theory and Case Studies*. Londýn,: Springer, 2019. 137 p. ISBN 978-3-030-17404.

Deadline for submission of Master thesis:	15. 05. 2026
Approval of assignment of Master thesis:	21. 04. 2026
Assignment of Master thesis approved by:	prof. Ing. Miroslav Fikar, DrSc. – Study programme supervisor